Integration CHEAT SHEET

THE INTEGRAL AND ITS RULES

$\int f(x)\,dx = F(x) + C \quad \text{means} \quad F'(x) = f(x)$

$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \qquad n \neq -1$

$\int \frac{1}{x}\,dx = \ln|x| + C$

$\int [k\,f(x) + g(x)]\,dx = k\int f(x)\,dx + \int g(x)\,dx$

check any answer: differentiate it, and the integrand must come back

STANDARD INTEGRALS (EACH + C)

f(x)	∫ f(x) dx
e^x	e^x
a^x	$a^x / \ln a$
$\sin x$	$-\cos x$
$\cos x$	$\sin x$
$\sec^2 x$	$\tan x$
$\operatorname{cosec}^2 x$	$-\cot x$
$\sec x \tan x$	$\sec x$
$\operatorname{cosec} x \cot x$	$-\operatorname{cosec} x$
$1 / \sqrt{(1 - x^2)}$	$\sin^{-1} x$
$1 / (1 + x^2)$	$\tan^{-1} x$

each line is a derivative read from right to left

TAN, COT, SEC, COSEC

f(x)	∫ f(x) dx
$\tan x$	$\ln\lvert\sec x\rvert$
$\cot x$	$\ln\lvert\sin x\rvert$
$\sec x$	$\ln\lvert\sec x + \tan x\rvert$
$\operatorname{cosec} x$	$\ln\lvert\operatorname{cosec} x - \cot x\rvert$

SIX STANDARD FORMS (EACH + C)

f(x)	∫ f(x) dx
$1 / (x^2 - a^2)$	$\frac{1}{2a} \ln\left\lvert\frac{x-a}{x+a}\right\rvert$
$1 / (a^2 - x^2)$	$\frac{1}{2a} \ln\left\lvert\frac{a+x}{a-x}\right\rvert$
$1 / (x^2 + a^2)$	$\frac{1}{a} \tan^{-1} \frac{x}{a}$
$1 / \sqrt{(a^2 - x^2)}$	$\sin^{-1} \frac{x}{a}$
$1 / \sqrt{(x^2 - a^2)}$	$\ln\lvert x + \sqrt{(x^2 - a^2)}\rvert$
$1 / \sqrt{(x^2 + a^2)}$	$\ln\lvert x + \sqrt{(x^2 + a^2)}\rvert$

$a^2 - x^2$ under a root → $\sin^{-1}$ · $x^2 + a^2$ → $\tan^{-1}$ · the rest → ln

COMPLETING THE SQUARE

$x^2 + bx + c = (x + \frac{b}{2})^2 + c - \frac{b^2}{4}$

e.g. $x^2 + 6x + 13 = (x + 3)^2 + 2^2$

SUBSTITUTION

$\int f(g(x))\,g'(x)\,dx = \int f(u)\,du, \quad u = g(x)$

$\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + C$

$\int f(ax + b)\,dx = \frac{1}{a} F(ax + b) + C$

choose u · find du · rewrite in u · integrate, put x back

IDENTITIES THAT HELP

$\sin^2 x = \frac{1 - \cos 2x}{2} \rightarrow \frac{x}{2} - \frac{\sin 2x}{4}$

$\cos^2 x = \frac{1 + \cos 2x}{2} \rightarrow \frac{x}{2} + \frac{\sin 2x}{4}$

$2 \sin A \cos B = \sin(A + B) + \sin(A - B)$

$\tan^2 x = \sec^2 x - 1 \rightarrow \tan x - x$

odd powers of sin: peel one off, put u = cos x

INTEGRATION BY PARTS

$\int u\,dv = uv - \int v\,du$

$\int e^x [f(x) + f'(x)]\,dx = e^x f(x) + C$

u by ILATE: inverse trig · log · algebraic · trig · exponential

PARTIAL FRACTIONS

factor in the bottom	pieces
$x - a$	$A / (x - a)$
$(x - a)^2$	$A / (x - a) + B / (x - a)^2$
$x^2 + bx + c$	$(Ax + B) / (x^2 + bx + c)$

$x^2 + bx + c$ with no real roots · top's degree ≥ bottom's: divide first

Integration CHEAT SHEET

THE DEFINITE INTEGRAL

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \Sigma\, f(x_i)\,\Delta x, \quad \Delta x = \frac{b-a}{n}$$

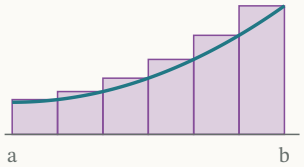
$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x)$$

$$\int_a^b f(x)\,dx = [F(x)]_a^b = F(b) - F(a)$$

$$\int_a^b u\,dv = [uv]_a^b - \int_a^b v\,du$$

the fundamental theorem: the area function's slope is f(x); evaluate as F(b) − F(a)

AREA AS A LIMIT OF STRIPS



strip: height $f(x_i)$, width Δx
more strips → the area
below the axis counts as −
C cancels in F(b) − F(a)
substitution: change the limits

WHY IT WORKS

$$A(x) = \int_a^x f(t)\,dt, \quad A'(x) = f(x)$$

$$F = A + C \;\rightarrow\; \int_a^b f(x)\,dx = F(b) - F(a)$$

FROM THE COURSE

$$\int_0^1 x^2\,dx = \frac{1}{3}$$

$$\int_0^{\pi} \sin x\,dx = 2$$

$$\int_1^e \frac{1}{x}\,dx = 1$$

$$\int_{-1}^2 |x|\,dx = \frac{5}{2}$$

$$\int_0^{10} 2t\,dt = 100 \quad \text{(the car)}$$

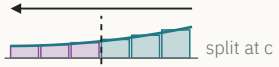
PROPERTIES

$$\int_a^a f(x)\,dx = 0$$

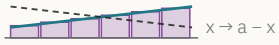
$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$$



$$\int_a^b f = \int_a^c f + \int_c^b f$$



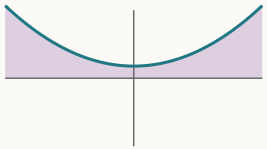
$$\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$$



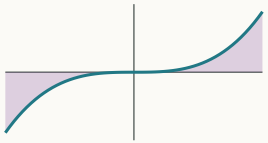
$$\int_{-a}^a f = 2\int_0^a f \quad \text{if } f \text{ is even}$$

$$\int_{-a}^a f = 0 \quad \text{if } f \text{ is odd}$$

EVEN AND ODD ON [−A, A]



even: twice one side



odd: the areas cancel

THE A − X MIRROR, USED

$$\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}}\,dx = \frac{\pi}{4}$$

replace x by π/2 − x: sine and cosine swap; add the two forms

AREAS

under a curve: $A = \int_a^b y\,dx \quad (y \geq 0)$

between curves: $A = \int_a^b (\text{top} - \text{bottom})\,dx$

sideways strips: $A = \int_c^d x\,dy$

find where curves meet; split where a curve crosses the axis

VOLUMES OF REVOLUTION

discs: $V = \int_a^b \pi\, y^2\,dx$

washers: $V = \int_a^b \pi(R^2 - r^2)\,dx$

about the y-axis: $V = \int_c^d \pi\, x^2\,dy$

cone $\frac{1}{3}\pi r^2 h$ · sphere $\frac{4}{3}\pi r^3$ (both from discs)

square each radius first: $R^2 - r^2$, not $(R - r)^2$

DIFFERENTIAL EQUATIONS

$$\frac{dy}{dx} = g(x)\,h(y) \;\rightarrow\; \int \frac{dy}{h(y)} = \int g(x)\,dx$$

$$\frac{dy}{dt} = ky \;\rightarrow\; y = y_0\, e^{kt}$$

$$\frac{dT}{dt} = -k(T - T_s) \;\rightarrow\; T = T_s + (T_0 - T_s)\, e^{-kt}$$

order: highest derivative · add C, then use the starting value